Callum Rhodes – CV

Please visit callum-rhodes.github.io for an interactive version of this CV

26 Bridgeman Way, Bristol, BS10 7GQ, UK
callum.r.rhodes@gmail.com
(+44)7751507545

Personal Statement

Callum Rhodes is a skilled robotics research engineer with experience across the spectrum of the field, from perception to control to embedded application. He has over 200 citations with papers at top international robotics conferences, as well as having studied at the prestigious Imperial robot vision lab. He is passionate about bringing theoretical solutions through to application and demonstration, which is shown via his many practical robotic showcases through both his PhD and his work at Frazer-Nash. He has also proven his leadership abilities by leading teams ranging from 4 to 20+ on a wide array of these projects.

Employment

2024 - 2026 **Frazer-Nash Consultancy – Lead Autonomy Engineer**

- Developed key autonomy functions for various UAV based projects which included target acquisition and full pose tracking, as well as predictive horizon motion modelling. Also lead development on autonomous control strategies using constrained optimisation methods to allow dynamically constrained platforms to achieve their missions.
- Tech lead for a £2m project that involved management and organisation of a multidisciplinary team of over 20 people. Oversaw both software and hardware implementation whilst developing cutting edge autonomy functions. Also successfully demonstrated the system at a live customer trial with key stakeholders present.

2023 - 2024 **Dyson Robotics Lab, Imperial College London – Research fellow**

- Under the supervision of Prof. Andrew Davison, developed distributed inference techniques for efficient computation of 3D vision problems, as well as leveraging monocular learnt priors for bootstrapping SLAM systems
- Projects include: Applying a dynamic multigrid structure to distributed inference techniques to speed up convergence, leveraging learned priors to achieve 3D normal integration with monocular images and estimating extrinsic camera rotation using structure in the environment.

2022 - 2023 **Frazer-Nash Consultancy – Control engineer**

- Worked on designing the perception and control systems of assured autonomous UAVs. Also developed new simulation tools to allow rapid prototyping of autonomous functions (which are still widely used within the company)
- Refactored old code bases and introduced source control to fix long term bugs and provide a better platform for further development

2019 - 2022 **Loughborough University – Research Associate**

- Alongside PhD studies, undertook the lead research role on a DSTL funded project investigating efficient source term estimation using unmanned aerial vehicles.
- Due to the quality of the research output, was able to secure several consecutive years of additional project funding to develop the source search system further.

Education

2018 – 2022 **Loughborough University – PhD Field Robotics**
- PhD candidate in autonomous systems. Researching methods for automating robotic first response systems for safe and efficient situational awareness of complex HAZMAT and disaster scenarios.
- Published in several International conferences and journals including: ICRA, IROS, RAL and T-ASE
- Won first place at the school's research day for the research output achieved over the course of the PhD

2013 - 2018 **Loughborough University – MEng Automotive Engineering**
- 1st class with hons
- Final Year Project developing an autonomous exploration system for mobile robotics

Extracurricular Events - Robotics

ELROB 2022
- Lead a team of researchers from Loughborough University in a European robotics trial aiming to test state of the art remote CRBN robotics tools for the acquisition of real-time radiation maps.
- Showcased PhD research and successfully identified the location of many radiation sources within an a-priori unknown and complex abandoned building.

EnRicH 2021
- Lead the first UK team to attend the European robotics hackathon located inside the Zwentendorf nuclear power plant, Austria. The challenge required semi-autonomous flight of a small UAV whilst mapping radiation signals through a 40m shaft.
- Achieved 2nd place in the UAV and mapping categories using only a 450mm vision only quadcopter.

Publications

Published in several top international conferences and journals including ICRA, IROS, RAL and 3DV. Please visit my Google scholar page for a full list of my publications: Scholar link

Skills/Qualifications Summary

Theoretical expertise	**Software skillset**	**Other skills**
Optimisation	Python	Project Management
Sensor fusion	MATLAB / Simulink	Team leader
3D vision	ROS 1/2	Customer relations
State estimation	C++	Full, clean car and motorbike licence
Model predictive control	Git	Basic Mandarin and Japanese Proficiency
Path planning	Linux	
SLAM	Docker	
Simulation	NX / Autodesk Fusion CAD	